Problem with Distribution ?

→ In many scenario, the number of outcomes can be much larger & hence a table would be tedious to write down.

worse still, the no. of possible outcomes could be infinite, in which case, good were writing a table.

Solution ? ⇒

What if we use a mathematical function to model the relationship between outcome and probability, which help to show the distribution graphically.

ex :- throwing 3 dice together.

x

Probability Distribution Function - PDF

A probability distribution function is a mathematical function that desc the probability of obtaining differe value of Random variable in a par probability distribution.

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Discrete R.V.

Continuous R.V.

Probability mass function (PMF)

Probability Density function (PDF)

$\downarrow$

Cumulative Probability

(running addition)

$\downarrow$ CMF

Cummulative Mass function

$\downarrow$ CDF

Cummulative Density funct

———— x ————

$\boxed{\text{PMF :- Probability Mass Function}}$

→ It is a mathematical function that describe the probability distribution a "Descreate Random variable".

Condition 1 :- The probability assign to each value must be non-negati

Condition 2 :- The sum of probability must be equal to 1.

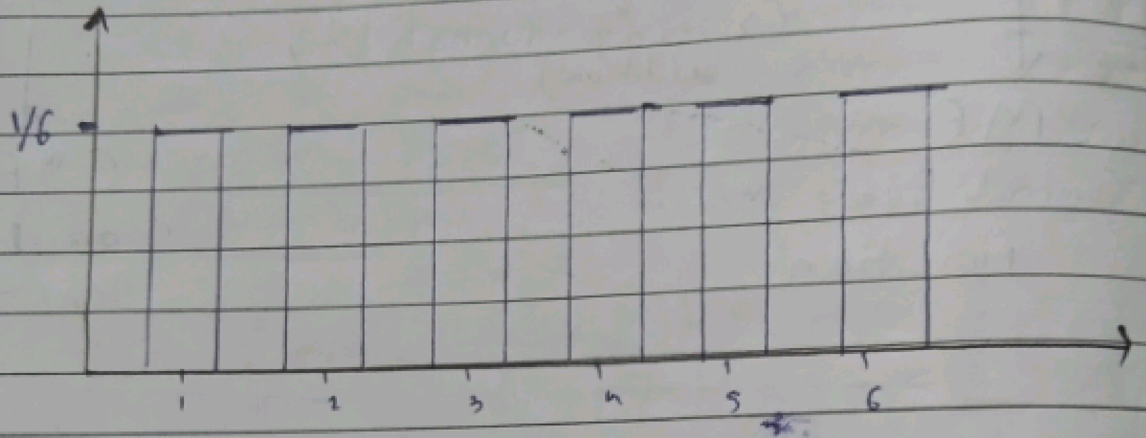
ex :- {1, 2, 3, 4, 5, 6} Rolling a Di

$P(1) = \frac{1}{6}$ $\quad P(2) = \frac{1}{6}$ - - -

Total Prob. $= P(1) + P(2) + P(3) + P(4) + P(\ldots$

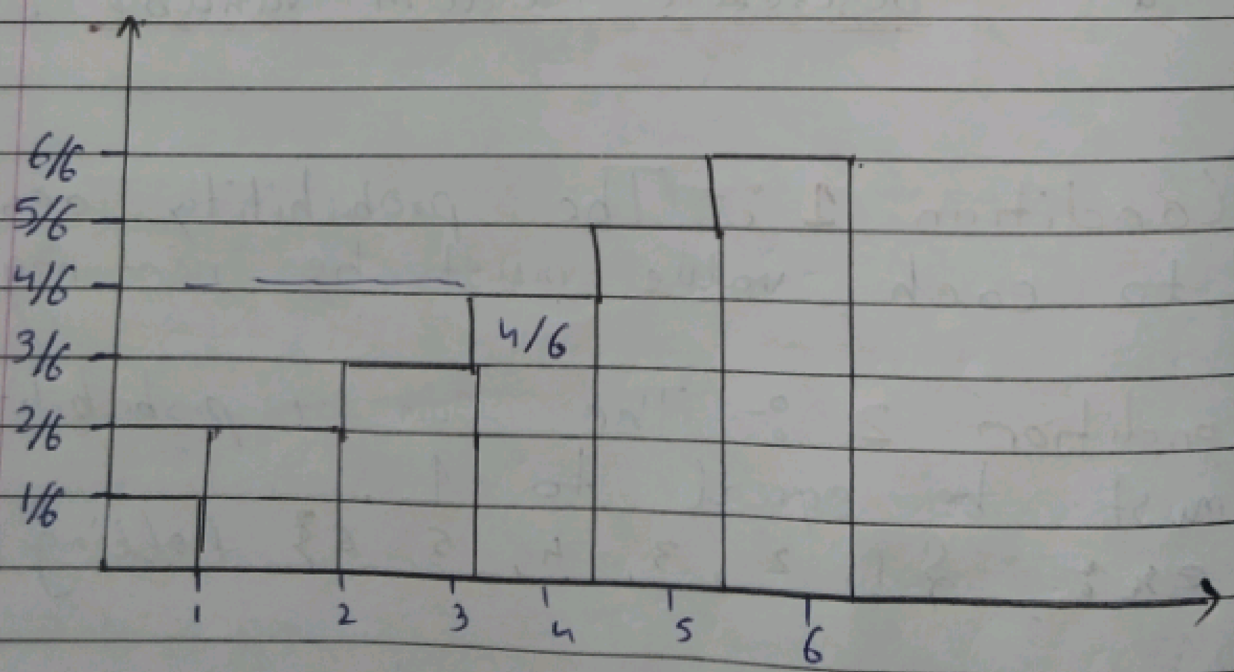
$= \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6}$

$= \frac{6}{6} = \boxed{1}$



CMF :- $\longrightarrow 6$

$P(x < 4) = P(1) + P(2) + P(3) + P(4)$



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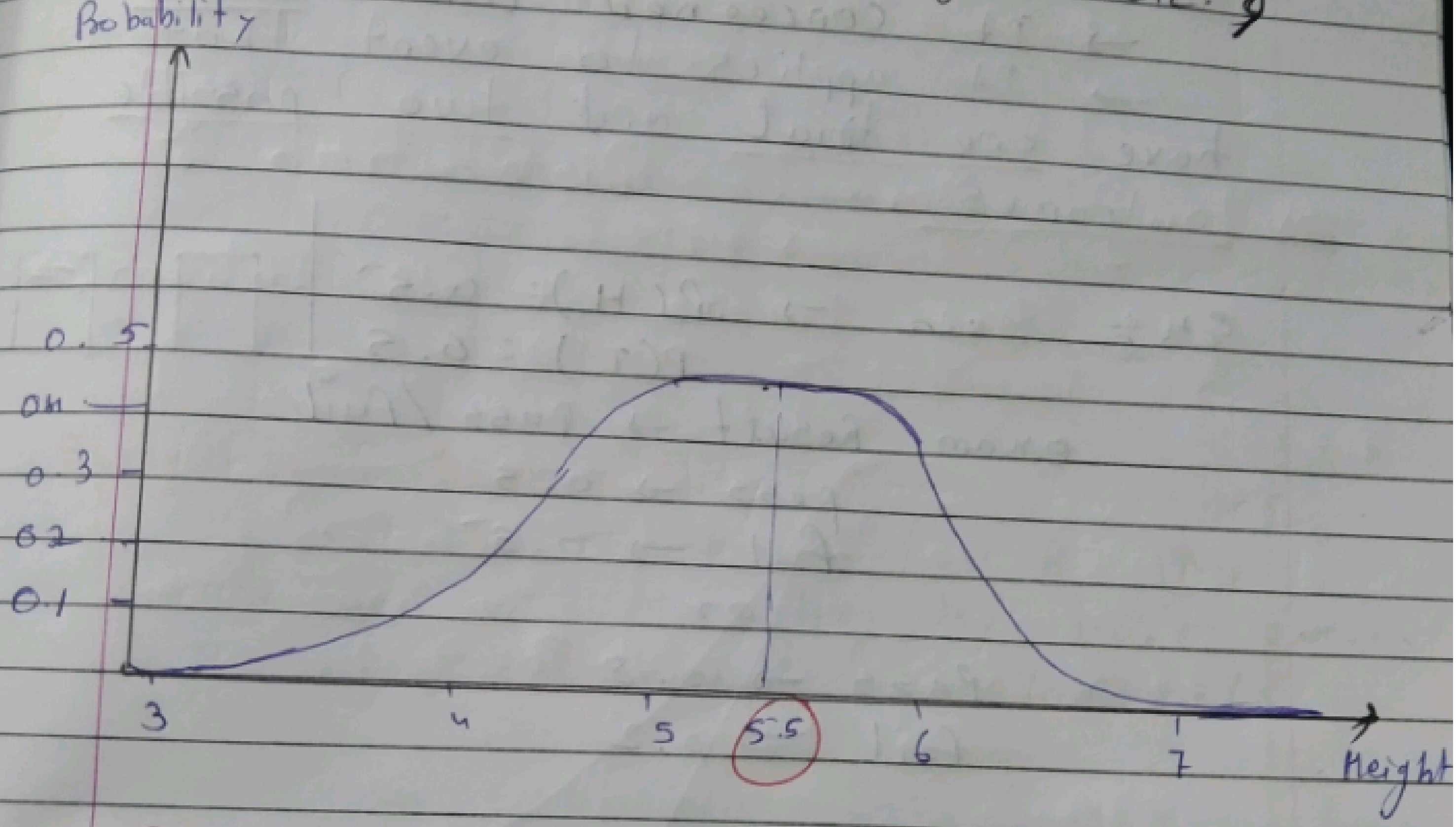
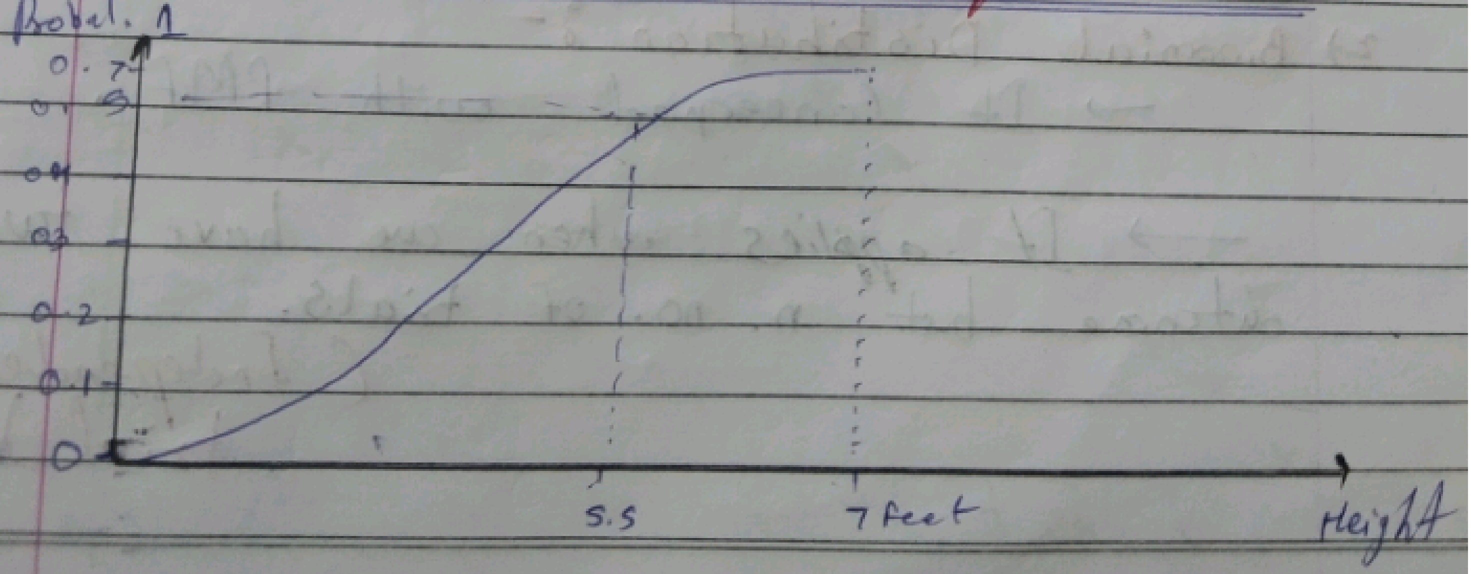
Probability Density function (PDF)

→ It is a mathematical function that descrete the probability distribution of a continious random variable.

ex:- { Heigh, weight, etc. }



CDF → Cumulative Density function



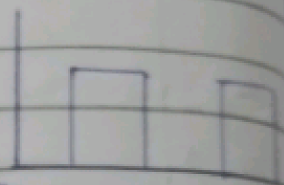
Types of Distribution (Probability Distrib...

1) Bernoulli Distribution :-

→ It is a discrete probability Distribution

→ It concerned with PMF

→ It applies to event that have one trial and two possible outcomes.



ex :- coin → $P(H) = 0.5$

$P(T) = 0.5$

exam Result → pass / fail

pass → 0.5

fail → 0.5

or

pass → 0.8

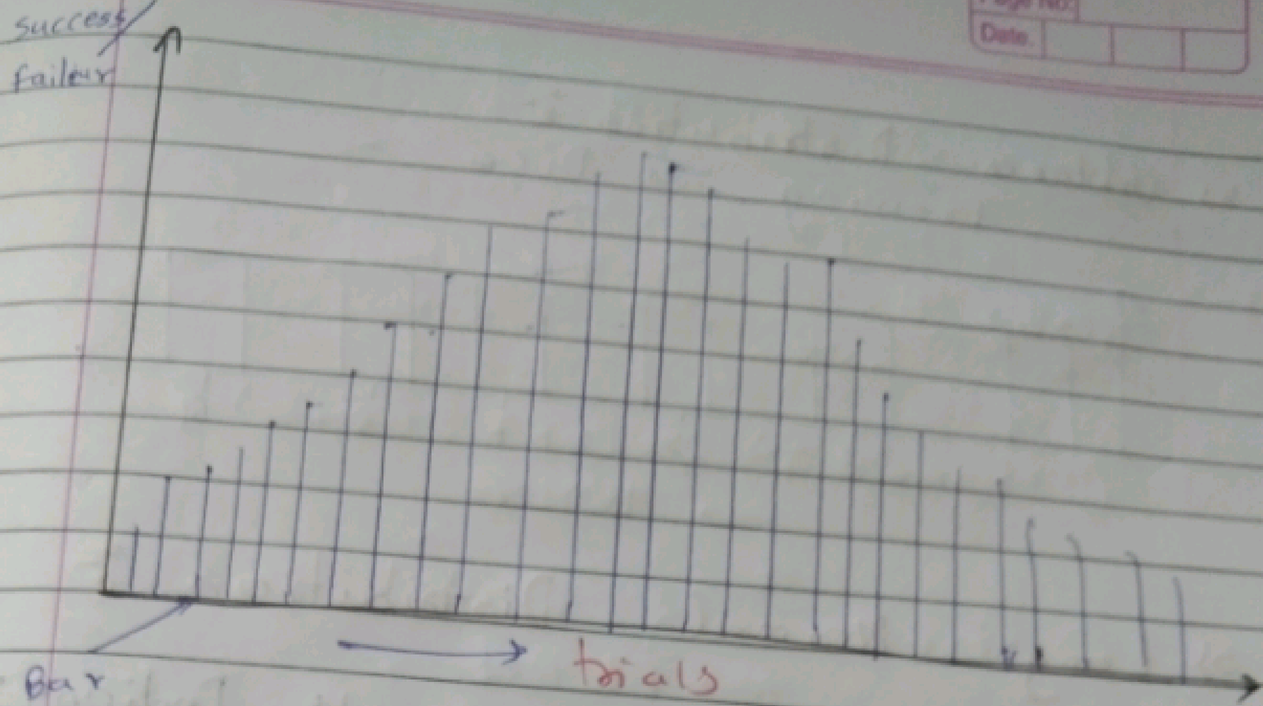
fail → 0.2

2) Binomial Distribution :-

→ It Concerned with PMF.

→ It applies when we have two outcome but n. no. of trials.

(Independent trial)

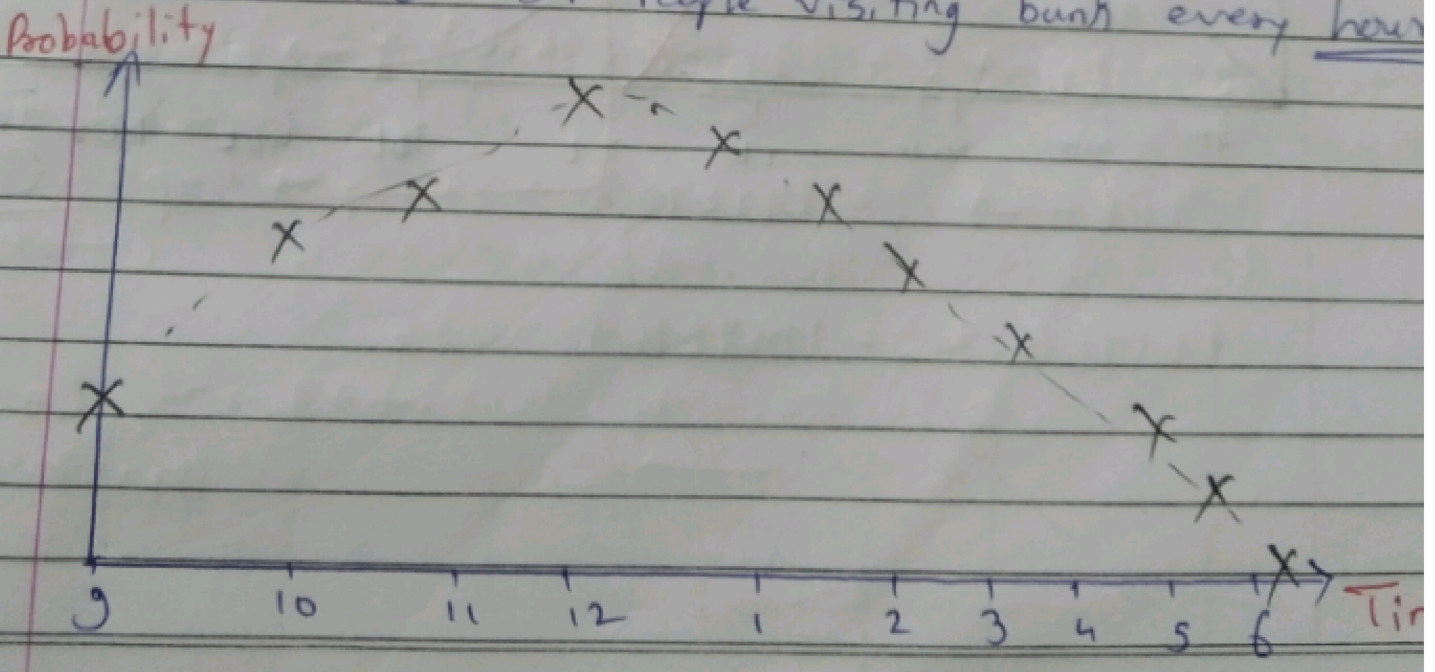


3) Poisson Distribution :-

→ It concerned with PMF

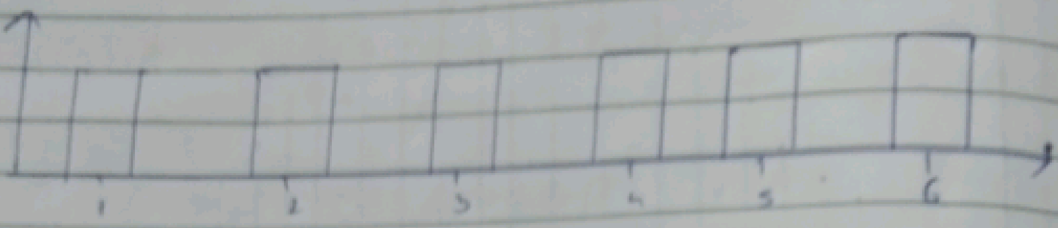
→ It applies when the no. of events occuring in a fixed time interval.

eg = no. of People visiting bank every hour



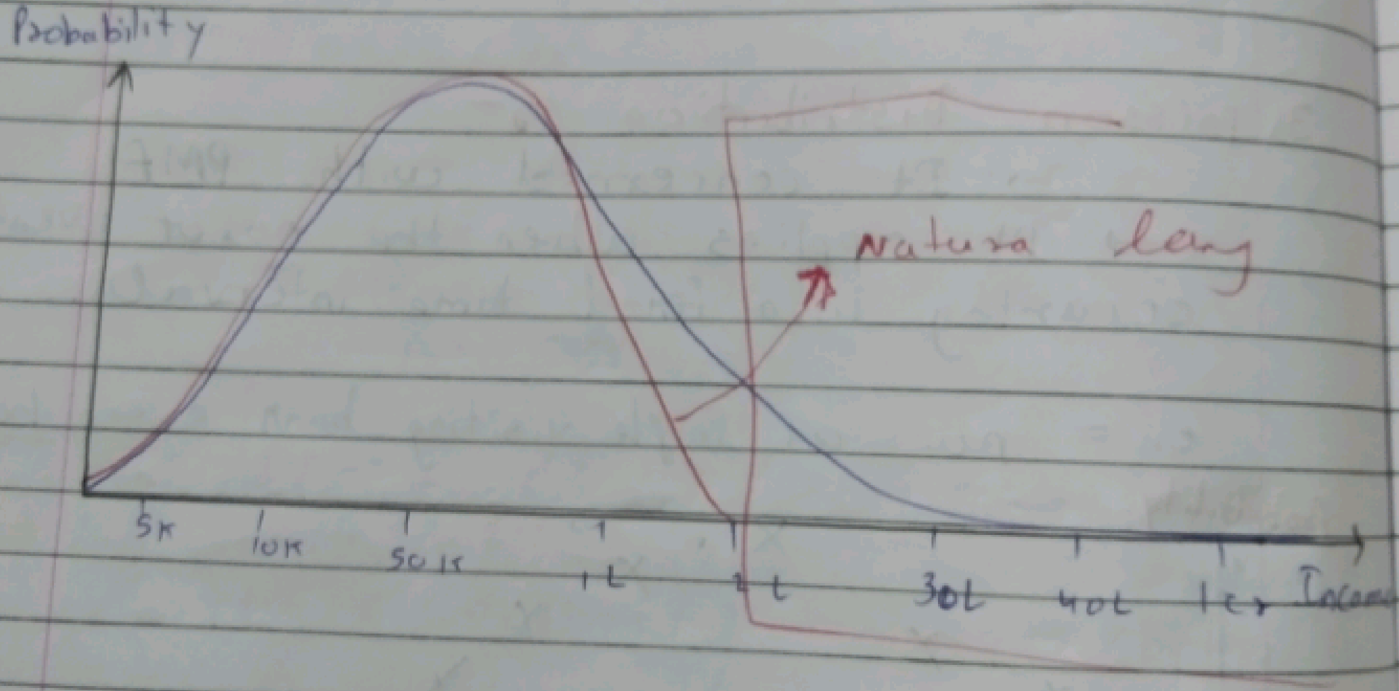
4) Uniform Distribution :-

Throwing a Dice -



5) Long Normal Distribution :-

ex:- Avg. Income of all Indians :-



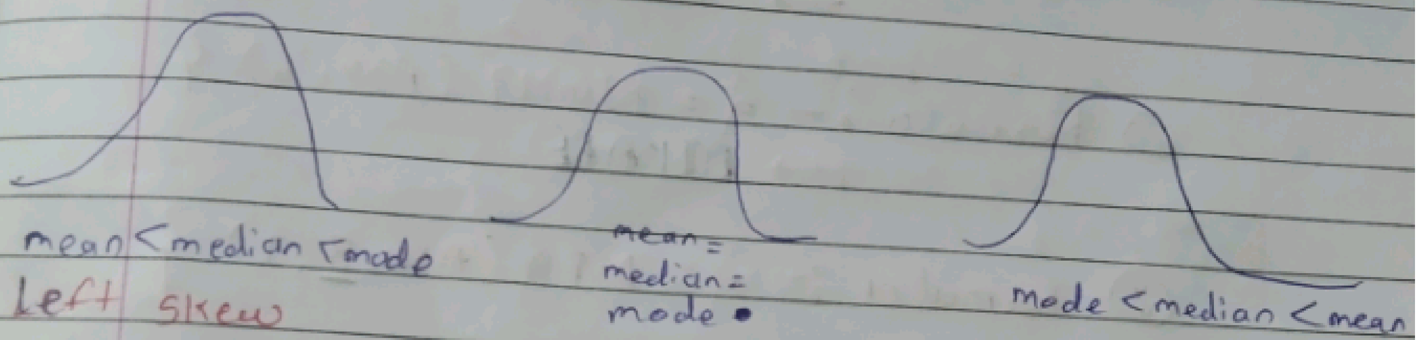
Normal / Guassian Distribution

↳ Continous RV (Height / weight / BP)

↳ It is systematical Distribution From the central tendency.

↳ It's look's like a Bell Curve

$\boxed{\text{mean} = \text{median} = \text{mode}}$



Skewness → [illegible]

Empirial Rule of Normal Distribution

68.3, 95.5 & 99.7% Rule

The imperial rule states that for a normal distribution .68.3% of the observati[on] data will lie inside one standard deviation of mean. 95.5% lies inside second SD. & 99.7% lies within thired SD.